MATH 110 — Week 7 — Solutions

Notes Chapters 12–13; Homework 7

HW7 P1.

Idea. Compute the first polynomial relation, then check whether its factors split and are distinct.

1. $A^2 = -I$, and A is not scalar over either field. Thus $m_A = x^2 + 1$, irreducible over $\mathbb{R}$, but $(x - i)(x + i)$ over $\mathbb{C}$. Hence A is diagonalizable only over $\mathbb{C}$.
2. $(B - I)^2 = 0$ but $B - I \neq 0$; B is nonscalar, so a degree-one relation is impossible. Thus $m_B = (x - 1)^2$, so B is not diagonalizable over $\mathbb{R}$.

HW7 P2.

Idea. The iterates of x^3 detect every possible lower-degree relation.

1. $D^4 = 0$, while $x^3, Dx^3, D^2x^3, D^3x^3 = x^3, 3x^2, 6x, 6$ are independent because their nonzero monomials have distinct degrees. Thus $m_D = x^4$.
2. Its repeated root rules out diagonalizability by Proposition 68.

HW7 P3.

Idea. A nonzero constant term allows the operator relation to be rearranged into an inverse.

1. If $a_0 = 0$, write $m_T = xq$; invertibility and $Tq(T) = 0$ imply $q(T) = 0$, contradicting minimality.
2. Writing $a_d = 1$, rearrangement gives $T^{-1} = -a_0^{-1} \sum_{j=1}^{d} a_j T^{j-1}$ for $d \geq 1$; for $d = 0$ on the zero space, $T^{-1} = I = 0$.
3. In (c), $m_T = (x-1)^2(x^2+3x+4)$. Thus $T^{-1} = (-T^3 - T^2 + T + 5I)/4$. Its eigenvalues are $1, (-3+i\sqrt{7})/2, (-3-i\sqrt{7})/2$, by Proposition 66 and the quadratic formula.
4. The repeated root 1 in the minimal polynomial rules out diagonalizability, by Proposition 68.

HW7 P4 (optional).

Idea. Affine substitution preserves annihilation and degree; reverse it to exclude smaller relations.

1. Put $S = T - \lambda I$ and $q(x) = m_T(x + \lambda)$. Then q is monic of the same degree as m_T and $q(S) = m_T(T) = 0$.
2. If a nonzero polynomial h of smaller degree annihilated S, then $h(x - \lambda)$ would annihilate T with smaller degree, impossible. Hence $m_S = q$.
3. For $R = cT$, $q(x) = c^d m_T(x/c)$ is monic of degree d and $q(R) = 0$. A smaller annihilator h of R would give the smaller annihilator $h(cx)$ of T. Thus $m_R = q$.

HW7 P5.

Idea. The root set and the degree bound leave just two monic possibilities.

1. Complex factorization and Proposition 66 imply $m_A = (x - \lambda)^d$. Since the space is nonzero, $d \geq 1$, and Proposition 67 gives $d \leq 2$.
2. The relation $m_A = x - \lambda$ is equivalent to $A - \lambda I = 0$. Proposition 68 gives diagonalizability exactly in this first case.

HW7 P6.

Idea. The repeated diagonal value has only one eigenvector direction.

1. Triangular form gives eigenvalues 2 and 3. $(C-2I)(x,y,z)=(y,0,z)=0$ gives $E(2,C)=Fe_1$; $(C-3I)(x,y,z)=(-x+y,-y,0)=0$ gives $E(3,C)=Fe_3$.
2. $(C-2I)(C-3I)=\left(\begin{smallmatrix}0&-1&0\\0&0&0\\0&0&0\end{smallmatrix}\right)\neq 0$, whereas $(C-2I)^2(C-3I)=0$. Thus $m_C=(x-2)^2(x-3)$.
3. The repeated root rules out diagonalizability. Equivalently the two eigenspace dimensions sum to 2, less than 3.

HW7 P7 (optional).

Idea. Interpolate the desired scalar action on each eigenspace.

1. Set $p_j(x)=\prod_{k\neq j}(x-\lambda_k)/(\lambda_j-\lambda_k)$. All denominators are nonzero, and substitution gives the required values.
2. Proposition 68 gives the direct decomposition $v=\sum_k v_k$ with $v_k\in E(\lambda_k,T)$. Week 6 evaluation on eigenvectors gives $p_j(T)v_k=p_j(\lambda_k)v_k$.
3. Therefore $p_j(T)v=v_j$, which is precisely the stated projection. It is identity on its range, zero on the other summands, and satisfies $p_j(T)^2=p_j(T)$.

Exam practice, Problem 6.

Idea. Use the same first-dependence and remainder arguments on one vector.

1. If $v=0$, take $p_v=1$; it is the unique monic degree-0 polynomial and divides every polynomial. Otherwise the finite-dimensional dependence of $v,Tv,\dots$ gives a nonzero annihilating polynomial for v. Choose minimal degree and normalize it to be monic.
2. Two distinct monic choices of the same degree would have nonzero lower-degree difference annihilating v, a contradiction; hence p_v is unique.
3. Divide $m_T=p_vh+r$ with $\deg r<\deg p_v$ or $r=0$. Evaluation on v gives $r(T)v=0$ because $p_v(T)$ commutes with $h(T)$. Minimality forces $r=0$, so $p_v\mid m_T$.

Actual Midterm 2, Question 3.

Idea. The two shifted kernels give the available eigenvectors. A quadratic product tests whether one factor per eigenvalue is enough; a cubic zero product supplies the upper bound.

1. The triangular diagonal gives $2,-1$. For $A-2I$, the equations are $y=0$, $-3z=0$, giving $E(2,T)=\mathbb{R}e_1$. For $A+I$, they are $3x+y=0$, $3y=0$, giving $E(-1,T)=\mathbb{R}e_3$.
2. m_A must contain both root factors. Multiplication gives $(A-2I)(A+I)=\left(\begin{smallmatrix}0&3&0\\0&0&0\\0&0&0\end{smallmatrix}\right)\neq 0$, but $(A-2I)^2(A+I)=0$. Thus $m_A=(x-2)^2(x+1)=x^3-3x^2+4$; no proper divisor containing both roots works.
3. The minimal polynomial has a repeated root, so T is not diagonalizable. Independently, its eigenspaces have dimensions $1+1<3$.

Practice Midterm 2, Question 3(a,b).

Idea. The range of $T-\lambda I$ is smaller and contains every difference $Tv-\lambda v$, so it controls the extra columns when a basis is extended.

1. In dimension 1, for any basis vector u there is a scalar a with $Tu=au$, giving a 1×1 triangular matrix. The empty basis handles dimension 0 when it occurs inside induction.
2. For $n>1$, choose an eigenvalue λ. The operator $T-\lambda I$ has a nonzero kernel; rank–nullity gives $\dim\operatorname{range}(T-\lambda I)<n$.
3. If $u=(T-\lambda I)v$, then $Tu=T(T-\lambda I)v=(T-\lambda I)Tv\in U$. This proves invariance.

Practice Midterm 2, Question 3(c,d).

Idea. A triangular basis inside U already handles its columns; the remaining columns need only lie in U plus their own line.

1. The restriction $T|_U$ is an operator on the smaller complex space U. Induction gives a basis u_j with $Tu_j \in \operatorname{span}(u_1, \dots, u_j)$; if $U = 0$, use its empty basis.
2. Extend it by $v_1, \dots, v_r$. For each j, $Tv_j = (T - \lambda I)v_j + \lambda v_j \in U + \mathbb{C}v_j$, contained in the span through that vector in the extended list.
3. Every basis image therefore belongs to its initial span. Proposition 59 makes the matrix upper triangular.

Practice Midterm 2, Question 4.

Idea. Rearrange the constant term of the minimal relation. If that term is zero, cancellation by an assumed inverse would contradict minimal degree.

1. Write $m_T = \sum_{j=0}^{d} a_j x^j$, $a_d = 1$. If $a_0 \neq 0$, then $T[-a_0^{-1} \sum_{j=1}^{d} a_j T^{j-1}] = I$. The bracket commutes with T, so it is a two-sided inverse.
2. If $a_0 = 0$, write $m_T = xq$. Invertibility would make $Tq(T) = 0$ imply $q(T) = 0$, contradicting the smaller degree of nonzero q.
3. $R^2 = -I$ by substitution. The map is nonscalar, so there is no degree-one relation and $m_R = x^2 + 1$.
4. From $R^2 + I = 0$, $R(-R) = (-R)R = I$. Hence $R^{-1} = -R$, namely $R^{-1}(x, y) = (y, -x)$, checked by both compositions.

Midterm 2 Review, Problem 4(1).

Idea. Differentiation couples each sine/cosine pair. Real eigenvalue equations on these pairs have no nonzero solution; only constants remain.

1. The listed functions are independent: their derivative-value or trigonometric coefficient equations separate the constant and the two sine/cosine pairs. More explicitly, they are eigenvectors of D^2 with values $0, -1, -4$ grouped by frequency; 1 is nonzero and each pair $\sin jx, \cos jx$ is independent by evaluating at 0 and $\pi/(2j)$. The distinct-eigenspace direct-sum argument from Week 6 therefore gives independence.
2. Write $f = a + b\sin x + c\cos x + d\sin 2x + e\cos 2x$. The equation $Df = \lambda f$ gives $\lambda a = 0$, $-c = \lambda b$, $b = \lambda c$, $-2e = \lambda d$, $2d = \lambda e$.
3. For real λ, these imply $(\lambda^2 + 1)b = (\lambda^2 + 1)c = 0$ and $(\lambda^2 + 4)d = (\lambda^2 + 4)e = 0$. Thus $b = c = d = e = 0$. A nonzero f requires $\lambda = 0$ and $a \neq 0$, so the only real eigenvalue is 0, with eigenspace the constants.
4. Its eigenspace dimension is $1 < 5$, so it is not diagonalizable over $\mathbb{R}$.

Midterm 2 Review, Problem 4(2,3).

Idea. The rotation's scalar equation is $\lambda^2 = -1$. A triangular basis would put such a scalar on its diagonal.

1. $T(x, y) = (y, -x)$ satisfies $T^2 = -I$. For $\lambda = i$, $(1, i)$ satisfies $T(1, i) = i(1, i)$; for $\lambda = -i$, $(1, -i)$ satisfies $T(1, -i) = -i(1, -i)$.
2. The two distinct eigenvalues give independent vectors, so $((1, i), (1, -i))$ is a complex eigenbasis and its matrix is $\operatorname{diag}(i, -i)$, in particular upper triangular.
3. Over $\mathbb{R}$, no scalar solves $\lambda^2 = -1$, so no real eigenvalue exists. Any nonempty real upper-triangular matrix has real diagonal eigenvalues by Proposition 60; hence no such basis exists.